EGOR DZHIN

AI Solutions Architect | Senior AI / ML Engineer

Senior AI and software engineering professional with experience applying **AI/ML, mathematical modeling, optimization, computer vision, data engineering, and LLM/RAG technologies** to industrial, healthcare, education, logistics, and enterprise systems.

Experience spans traditional and modern AI: industrial optimization and predictive modeling, deep learning, medical image processing, large-scale intelligent data processing, and production LLM/RAG systems.

Strong engineering background enables development of complete AI solutions across:

Data → Modeling → AI/ML → Backend Services → Integration → Deployment → Monitoring

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CORE AI / ML SKILLS

AI / Machine Learning

Python · PyTorch · scikit-learn · NumPy · Pandas · Hugging Face · Transformers · MLflow · Deep Learning · Machine Learning · Mathematical Modeling · Numerical Methods · Optimization · Stochastic Gradient Descent · ALOPEX · Hill Climbing

Generative AI / LLM

LLM · RAG · LangChain · PydanticAI · Ragas · AI Agents · GPT · Claude · Gemini · Prompt Engineering · Structured LLM Output · LLM Guardrails · LLM Evaluation

Retrieval / RAG

Retrieval-Augmented Generation · Parent-Child Chunking · Hybrid Search · Reranking · Context Retrieval · Retrieval Evaluation · Hallucination Reduction

Computer Vision / Medical AI

OpenCV · Image Processing · Object Tracking · Ultrasound Image Processing · Medical AI

AI Data Engineering

PostgreSQL · Oracle · MongoDB · SQL · ETL · Data Validation · Data Quality · Data Deduplication · Data Normalization · OLAP · Kafka · Airflow

AI Platform / Backend

Python · FastAPI · REST APIs · Microservices · Docker · Kubernetes · Azure · Google Cloud · CI/CD · Prometheus · Grafana

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SELECTED PROFESSIONAL EXPERIENCE

PMI Thailand Chapter

Senior Software Engineering Manager

May 2024 – Jun 2026 | Thailand

Led and contributed to AI-enabled software platforms combining LLM/RAG, backend engineering, data processing, and cloud-native architecture.

Production RAG / Customer Inquiry System

Designed and implemented a production RAG-based question-answering system for automating customer inquiries.

Architecture included:

Documents → Parent-Child Chunking → Hybrid Search → Reranking → Context Selection → LLM → Guardrails → Evaluation

Key work:

- Designed Parent-Child document chunking strategy.
- Implemented Hybrid Search for improved retrieval.
- Applied reranking to improve context relevance.
- Designed strict prompting and output guardrails to reduce hallucinations.
- Applied Ragas for systematic RAG evaluation.
- Optimized the system for accurate, low-latency responses.
- Integrated AI capabilities into production web/backend infrastructure.

Technologies: Python, RAG, LLM, LangChain, PydanticAI, Ragas, PostgreSQL, React, Prometheus, Grafana, Docker, Kubernetes

National Educational Information System — Thailand

Contributed to modernization of a national-level educational information environment involving heterogeneous educational data sources.

Worked on:

- Data staging and validation
- Deduplication
- Data standardization
- Standardized identifiers
- Centralized data integration
- Business Intelligence
- Educational data standards
- AI-assisted data processing

Applied **PydanticAI** to structured data-processing workflows.

The broader architecture included REST APIs, microservices, Kubernetes, Docker, React/Angular, Spring/.NET, and cloud-native services.

Thai Software Engineering Co., Ltd.

Senior Software Engineering Manager

Aug 2021 – Mar 2024 | Thailand

Worked on AI, mathematical modeling, mathematical algorithms, medical imaging, and large-scale data-processing platforms.

AI Ultrasound Image Processing

Worked on AI-based ultrasonic image-processing platforms supporting applications related to:

- Breast cancer diagnosis
- Thyroid cancer diagnosis

Responsibilities included:

- Mathematical modeling
- AI/image-processing system development
- Medical image processing
- Software/system architecture
- Full-stack integration
- Engineering leadership

Also worked with **HealthTAG — Trusted Health Data Infrastructure for Thailand**.

Integrated ISO 9001 quality-management principles into the software-development lifecycle.

Relevant technologies: AI/ML, mathematical modeling, image processing, REST APIs, microservices, Kubernetes, Docker, Golang

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New Computer Technology Consulting

Senior Software Engineer

Mar 2018 – Jul 2021

Laem Chabang Port Upgrade — AI/Data Engineering

Worked on a major port modernization and automation environment combining large-scale data processing, enterprise systems, industrial automation, networking, and AI-assisted data engineering.

Designed/worked with high-performance data architecture intended to support **billions of records** in a major deep-sea terminal environment handling more than **11 million TEUs annually**.

Worked with approximately **20 years of heterogeneous historical port data**, including:

- EDI messages
- Container information
- Shipping information
- Bills of lading
- Customs-related records
- Operator-entered logs
- PDFs
- Legacy databases/files

Designed distributed processing concepts involving:

Legacy Data → Temporal Chunking → Parallel Processing → Structured Extraction → Validation → PostgreSQL / OLAP

The architecture included AI-assisted structured-data processing using PydanticAI.

The broader port project also involved:

- 5G-enabled automation
- PLC systems
- High-availability interfaces

- Network protocols
- Enterprise/warehouse systems

SELECTED EARLIER APPLIED AI EXPERIENCE

Novartis Thailand

Project Manager / Senior Software Developer / Senior Backend Developer

2016 – 2017

Worked across pharmaceutical manufacturing IT, enterprise software, industrial automation, and applied AI/ML.

AI/ML Production Planning

Designed an AI/ML-based approach for estimating production quantities and supporting manufacturing planning.

Model inputs included:

- Historical production
- Orders and demand
- Inventory
- Production capacity
- Process time
- Raw-material availability
- Material-market information
- Raw-material price changes
- Production-cycle performance

Applied approaches included:

- Deep Learning
- ALOPEX
- Stochastic Gradient Descent

Also worked with ERP, SCADA, fieldbus deployment, automatic control systems, GMP, ISO 9001, and pharmaceutical production processes.

SJF Farm Reconstruction / Modernization

Project Manager / Senior Full-Stack Developer

2017 – 2019

Developed systems connecting enterprise software, production management, AI/optimization, and field automation.

AI/Optimization for Compound Feed

Designed an optimization approach for determining compound-feed recipes using **Hill Climbing**.

Inputs included:

- Historical data
- Production data
- Existing feeding recipes
- Production plans
- Raw-material prices
- Production prices
- Market information

Integrated:

ERP → Production Management → Feed Optimization → Field Automatic Control

Also developed mobile production-management applications and integrated NFC-based identification.

Hongsa Power / Industrial AI Project

Senior Developer

Lao IT Dev period

Worked on power-plant/turbine-related software and industrial AI.

Turbine Control Optimization

Applied AI/ML to historical turbine operational data to support fine-tuning and optimization of turbine speed/control behavior.

Integrated field sensor information into the software/control environment using **ADAM-series industrial I/O/data-acquisition modules**.

Architecture included:

Field Sensors → ADAM I/O → Control / SCADA → Operational Data → Historical Analysis → AI/ML Optimization

This work provided early practical experience combining physical industrial systems, operational data, control software, and machine learning.

SELECTED AI / R&D PROJECT

Embedded Robotics / Computer Vision

Designed and developed a robotic vehicle using **Banana Pi**.

Integrated:

- Three simultaneously controlled servo motors
- Robotic vehicle control
- IMU-based pose/orientation sensing
- GPS positioning
- Camera input
- OpenCV
- Object tracking

Combined embedded control, sensing, computer vision, positioning, and robotic actuation.

This project established a long-term technical interest in **autonomous and unmanned systems**.

DATABASE / AI SYSTEM PERFORMANCE EXPERIENCE

10M+ Record System Redesign

2019 – 2020

Investigated a severely constrained record-management system containing more than **10 million records** and growing by approximately **200,000 records per year**.

The existing processing architecture used approximately 10 low-cost computers and required **more than one week** for a complete processing cycle.

Determined that the fundamental bottleneck was software/database architecture rather than hardware.

Redesigned:

- Database access
- Query processing
- Processing logic
- N+1 query patterns
- Large-record-set processing

Result

Before: ~10 computers / >1 week
After: 1 computer / ~10 minutes

The optimized system operated on an existing **Core 2 Duo / 4 GB RAM** computer without requiring hardware replacement and maintained required operational response times below approximately **10 seconds**.

This project established a strong practical foundation in database processing and performance-oriented architecture.

EDUCATION

Bangkok University International College

Bangkok, Thailand | Aug 2008 – May 2014

Master of Engineering — Electrical & Computer Engineering

Final approximately two years of study.

Academic foundation included:

- Mathematical Modeling
- Numerical Analysis
- Computational Mathematics
- Differential Equations
- Mathematical Analysis
- Discrete Mathematics
- Information Theory
- MATLAB
- Computer Architecture
- Operating Systems
- C++
- Java
- Network Design
- Cybersecurity
- Cryptography

Achieved the **highest possible grade across university subjects**.

ADDITIONAL TECHNICAL FOUNDATION

Languages: Python, C/C++, TypeScript, JavaScript, Golang, Java, PHP, C#

Databases: PostgreSQL, Oracle, MongoDB, SQL, OLAP

Architecture: Microservices, REST APIs, Distributed Systems, Cloud-Native Architecture, Data Architecture, System Integration

DevOps: Docker, Kubernetes, CI/CD, Jenkins, Prometheus, Grafana, Azure, Google Cloud

Engineering Process: Agile, Scrum, DevOps, QA, QA Automation, ISO 9001

AI DOMAIN EXPERIENCE

- Generative AI / LLM
- Retrieval-Augmented Generation
- Medical AI
- Medical Image Processing
- Industrial AI
- Manufacturing Optimization
- Mathematical Modeling
- Computer Vision
- Intelligent Data Processing
- Robotics / Unmanned Systems
- Educational Data Systems
- Logistics / Port Data Systems